

12.60

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October 12, 2025

Question

A square matrix $\mathbf{B}$ is skew-symmetric if

(a) $\mathbf{B}^T = -\mathbf{B}$

(b) $\mathbf{B}^T = \mathbf{B}$

(c) $\mathbf{B}^{-1} = \mathbf{B}$

(d) $\mathbf{B}^{-1} = \mathbf{B}^T$

Solution

B is a skew-symmetric, then by quadratic property

$$\mathbf{x}^\top \mathbf{B} \mathbf{x} = 0 \quad \forall \mathbf{x} \in \mathbb{R}^n \quad (1)$$

Consider two vectors $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$:

$$(\mathbf{x} + \mathbf{y})^\top \mathbf{B}(\mathbf{x} + \mathbf{y}) = \mathbf{x}^\top \mathbf{B} \mathbf{x} + \mathbf{x}^\top \mathbf{B} \mathbf{y} + \mathbf{y}^\top \mathbf{B} \mathbf{x} + \mathbf{y}^\top \mathbf{B} \mathbf{y} \quad (2)$$

$$\text{Since } \mathbf{x}^\top \mathbf{B} \mathbf{x} = \mathbf{y}^\top \mathbf{B} \mathbf{y} = 0, \text{ we have:} \quad (3)$$

$$\mathbf{x}^\top \mathbf{B} \mathbf{y} + \mathbf{y}^\top \mathbf{B} \mathbf{x} = 0 \quad \Rightarrow \quad \mathbf{x}^\top \mathbf{B} \mathbf{y} = -\mathbf{y}^\top \mathbf{B} \mathbf{x} \quad (4)$$

$$\text{Take } \mathbf{x} = \mathbf{e}_1, \mathbf{y} = \mathbf{e}_2 \text{ (standard basis vectors)} \quad (5)$$

$$\mathbf{e}_1^\top \mathbf{B} \mathbf{e}_2 = -\mathbf{e}_2^\top \mathbf{B} \mathbf{e}_1 \quad \Rightarrow \quad b_{ij} = -b_{ji} \quad (6)$$

$$\text{Since this holds for all } i, j, \text{ we conclude:} \quad (7)$$

$$\boxed{\mathbf{B}^\top = -\mathbf{B}} \quad (8)$$

Example:

$$\mathbf{B} = \begin{pmatrix} 0 & 2 & -1 \\ -2 & 0 & 3 \\ 1 & -3 & 0 \end{pmatrix}$$

$$\mathbf{B}^T = \begin{pmatrix} 0 & -2 & 1 \\ 2 & 0 & -3 \\ -1 & 3 & 0 \end{pmatrix} = -\mathbf{B}$$

C Code

```
#include <stdio.h>

#define N 3 // size of the square matrix

int main() {
    int B[N][N] = {
        {0, 2, -1},
        {-2, 0, 3},
        {1, -3, 0}
    };

    int isSkewSymmetric = 1; // 1 = true, 0 = false

    // Check the transpose property  $B^T = -B$ 
    for (int i = 0; i < N; i++) {
        for (int j = 0; j < N; j++) {
            if (B[i][j] != -B[j][i]) {
                isSkewSymmetric = 0;
            }
        }
    }
}
```

```
        break;
    }
}
if (!isSkewSymmetric) break;
}
printf(Matrix B:\n);
for (int i = 0; i < N; i++) {
    for (int j = 0; j < N; j++) {
        printf(%4d, B[i][j]);
    }
    printf(\n);
}
printf(\nIs B skew-symmetric ( $B^T = -B$ )? %d\n,
    isSkewSymmetric);

return 0;
}
```

```
import numpy as np

# Define a skew-symmetric matrix B
B = np.array([[0, 2, -1],
              [-2, 0, 3],
              [1, -3, 0]])

# Compute the transpose
B_T = B.T

# Compute -B
neg_B = -B
```

```
# Check if transpose equals negative of B
is_skew_symmetric = np.allclose(B_T, neg_B)

print(Matrix B:)
print(B)
print(\nTranspose B^T:)
print(B_T)
print(\nNegative of B (-B):)
print(neg_B)
print(\nIs B skew-symmetric (B^T = -B)?, is_skew_symmetric)
```